FĂTU MIRUNA MARIA
311 CA

TERMISTORUL

TABEL DE VALORI

$t(^\circ C)$	$R(\Omega)$		$T(K)$	$\frac{1}{T}$ $(10^{-3} K^{-1})$	$\ln R$	
	încălzire	răcire			↑	↓
40	333	321	313	318	5,80	5,77
45	292	274	318	314	5,67	5,61
50	242	234	323	309	5,48	5,45
55	215	200	328	304	5,37	5,29
60	186	171	333	300	5,22	5,14
65	161	146	338	295	5,08	4,98
70	138	125	343	291	4,92	4,82
75	118	106	348	287	4,77	4,66
80	102	92	353	283	4,62	4,52
85	89	80	358	279	4,48	4,38
90	78	69	363	275	4,35	4,23
95	67	60	368	271	4,20	4,09
100	58	53	373	268	4,06	3,97

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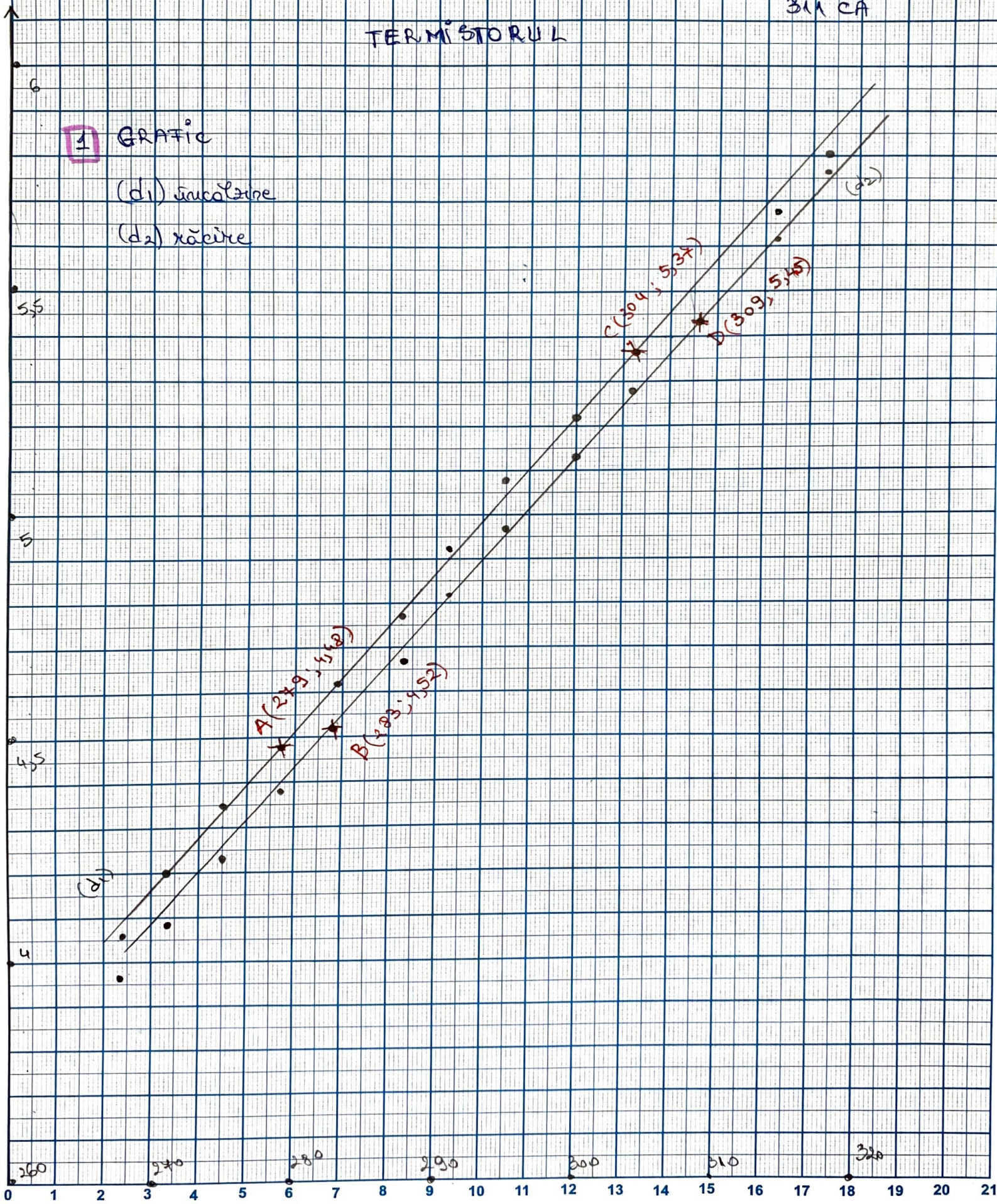
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1 GRAFIC

(d_1) încălzire

(d_2) răcire



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2 PANTELE DREPTELOR

→ Panta de încălzire

$$m_i = \frac{\Delta y}{\Delta x} = \frac{\ln R_C - \ln R_A}{\frac{1}{T_C} - \frac{1}{T_A}} = \frac{5,37 - 4,48}{(304 - 279)\frac{10^{-5}}{K}} = \frac{0,89}{25 \cdot 10^{-5}} K = 3560\,K$$

→ Panta de răcire

$$m_r = \frac{\Delta y}{\Delta x} = \frac{\ln R_D - \ln R_B}{\frac{1}{T_D} - \frac{1}{T_B}} = \frac{5,45 - 4,52}{(309 - 283)\frac{10^{-5}}{K}} = \frac{0,93}{26 \cdot 10^{-5}} K = 3576\,K$$

3

$$m_m = \frac{m_1 + m_2}{2} = \frac{3560 + 3576}{2} = 3568\,K$$

$$\ln R = c + \frac{\Delta E}{2kT} \Rightarrow \ln R \simeq \frac{\Delta E}{2kT} \rightarrow \ln R \cdot T \propto \frac{\Delta E}{2k} \Rightarrow \frac{\ln R}{\frac{1}{T}} \propto \frac{\Delta E}{2k}$$

$$\Rightarrow \frac{\Delta E}{2k} \propto m \Rightarrow \Delta E = 2km, \quad k = 8,6 \cdot 10^{-5} \frac{eV}{K}$$

$$\Delta E \propto 2 \cdot 8,6 \cdot 10^{-5} \cdot \frac{eV}{K} \cdot 3576K \propto 0,615\,eV$$ (lărgimea benzii interzise)

↳ $\boxed{\Delta E \propto 0,615 eV}$

SAU

$\Delta E \propto 2km$

La ÎNCĂLZIRE: $\Delta E_1 \propto 2km_1 \Rightarrow \Delta E_1 \propto 2 \cdot 8,6 \cdot 10^{-5} \frac{eV}{K} \cdot 3560K \simeq$

$\simeq 0,612\,eV$

La RĂCIRE: $\Delta E_2 \simeq 2km_2 \Rightarrow \Delta E_2 \simeq 2 \cdot 8,6 \cdot 10^{-5} \frac{eV}{K} \cdot 3576K \simeq$

$\simeq 0,615\,eV$

$$\Delta E = \frac{\Delta E_1 + \Delta E_2}{0,615} \propto \Rightarrow \Delta E \simeq \frac{0,612 + 0,615}{0,615\,2} \propto 0,613 eV$$

$\simeq 0,615\,eV$